FN-PIOTR-CASE2-001

Can the controller show each record was NOT accessed through the exposed key?

Prepared for	Piotr Reder — case 2 (synthetic): 'No entry is a claim, not a default'
Reconstruction ID	`FN-8c66ed94f110fa24`
Engine version	camsva 0.9.0
Generated	2026-10-21
Status	**FINAL — BOUNDED**

Purpose & boundary

This states, per proposition, what the records **establish**, **contradict**, or **cannot establish**.

STRICTLY NON-DISPOSITIVE. This synthetic case examines whether the controller can substantiate NON-ACCESS to particular records following exposure of an API key. It does NOT determine whether the incident constitutes a personal-data breach under GDPR Art. 4(12) — which also covers destruction, loss, alteration and unauthorised disclosure, not only access — nor whether notification obligations under Art. 33 arise; those assessments require a broader factual and legal evaluation, per breach and per data subject, by the controller and its supervisory authority. What this reconstruction feeds is the evidentiary question upstream of that: a record whose non-access cannot be established should not be classified as unaffected solely on the basis of the submitted access logs. Any consequential classification remains with the accountable controller. FactNotebook states only what the surviving records permit one to establish about access, per record and per path; the judgment and any consequential decision stay with the accountable authority. EVERY record is tested at TWO LEVELS (the symmetry Piotr Reder asked for): what the submitted log RECORDS, and what that does NOT establish about the world. At the RECORD level the log says a great deal — four records are ESTABLISHED silences (R-A/B/C/D-record: no read attributed to the key is recorded) and one is CONTRADICTED (R-E-record: the log contains a read of I-3301 attributed to the key). At the WORLD level NONE of it establishes real-world non-access: all five R-*-world propositions are CANNOT ESTABLISH, each for its named reason, and R-E-world too (the recorded read is attributed by DECLARATION, not independently verified). That gap between a rich record level and an empty world level IS the finding. Attribution rule (made explicit, not left implicit): attribution is declared for EVERY record — R-E's recorded read no less than R-A's absence of one — and the finding does not silently hold the two to different standards. A controller may, as a precautionary DECISION, treat a recorded-but-declared read as potentially affected, because the burden sits with the party asserting 'unaffected' (Art. 5(2) / 33(5)) and over-inclusion is the safer error; but that is a decision rule, NOT evidentiary truth, and it does not change what the records ESTABLISH. Every state here (ESTABLISHED / CONTRADICTED / CANNOT ESTABLISH) is relative to the admitted observation set; CONTRADICTED on R-E-record is not an independent certification that the access occurred in the world, only that the controller's own admitted record places a read there. The per-path non-access propositions cannot be established, each for a DIFFERENT, named reason, and the strongest (R-A) is DECLARED, not supported: its logging structure is present, but the anchor remains simulated AND complete capture of relevant accesses through the specified path is not demonstrated — and even were those resolved, it would reach only non-access via the logged path, not across every path. 'Not accessed' is a claim the records must affirmatively support — not a default that holds whenever a log happens to be silent. No Exempt Node: FactNotebook reconstructs from the controller's own bundle; it does not exempt itself, and it does not certify the controller's logging as adequate. A note on states: the engine reports three — ESTABLISHED / CONTRADICTED / CANNOT ESTABLISH. R-A is CANNOT ESTABLISH like R-B/C/D, but for a categorically different reason: the logging structure is present, but the anchor remains simulated and complete capture of relevant accesses through the specified path is not demonstrated. It is labelled DECLARED to mark that — an interpretable overlay on the machine state, not a fourth engine state. The distinct reason each record stops (coverage / object / integrity-structure / declared-anchor / path-completeness) is a RECORDED per-proposition field — carried in the machine-readable finding and in the Results table's 'determining limitation' column, not only in prose — declared as a characterisation of the observations and kept OUTSIDE the three engine states (ESTABLISHED / CONTRADICTED / CANNOT ESTABLISH) rather than collapsed into them, so a

reader of the table alone still sees why each world-level row cannot be established.

Provenance & processing

One integrity-verified synthetic bundle. The MANIFEST.sha256 (over eight evidence files plus assess.py, build_evidence.py and README.md — eleven in all) was recomputed here and matched 11/11; every driving fact was then separately re-derived from the submitted files, without relying on the conclusions of assess.py (not an independent verification of how the logs themselves were produced). Matching digests show the bundle is internally consistent WITH THE SUBMITTED MANIFEST — not, on its own, the origin of the files nor their state before the bundle was constituted. That consistency is distinct again from the records being COMPLETE or INDEPENDENT of the controller. All three — pre-constitution integrity, completeness and independence — are declared, not established, and the finding treats them as such.

Scope & coverage

Records were constituted and supplied by the controller (synthetic) — Piotr Reder's case-2 pack, submitted to FactNotebook as one bundle of logs, heartbeats, anchors, config and incident metadata, and admitted without selective exclusion of candidate observations — all extracted candidates were admitted. This finding is bounded to that submitted surface; the absence of internal selection does not establish the submission's completeness relative to the represented system, nor whether an adverse class of observations was withheld — that completeness is not established.

See Annex C (acquisition / selection boundary) and Annex B (non-admitted candidate register).

Results

Every tested proposition with its engine state and, where one applies, the determining limitation. Full detail follows below.

Proposition	State	Determining limitation
R-A-record — no read of orders record O-0417 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	the submitted log contains no read of O-0417 attributed to the key (documentary)
R-B-record — no read of invoices record I-0988 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	the submitted log contains no read of I-0988 attributed to the key (documentary)
R-C-record — no row-level read of customers record C-5521 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	no row-level read of C-5521 recorded — only a call-level query attributed to the key
R-D-record — no read of addresses record A-7730 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	the submitted log contains no read of A-7730 attributed to the key (documentary)
R-E-record — no read of invoices record I-3301 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**CONTRADICTED**	a read of I-3301 attributed to the key (declared) IS recorded at 03:05 — contradicts 'the log is silent'; record-level only
R-A-world — orders record O-0417 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)	**CANNOT ESTABLISH**	DECLARED, not supported — structure present, but anchor only simulated and capture not demonstrated (verified anchor + attribution are necessary, not sufficient)

R-B-world — invoices record I-0988 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)	**CANNOT ESTABLISH**	coverage gap — logger-invoices silent 02:09–02:55 (45 beats absent)
R-C-world — customers record C-5521 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)	**CANNOT ESTABLISH**	object granularity — call-level logging over a query that matches the record
R-D-world — addresses record A-7730 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)	**CANNOT ESTABLISH**	integrity structure absent — log unchained, store writable by db-admin
R-E-world — invoices record I-3301 was not accessed through key k_7f3a during the exposure window (real access, not merely the recorded entry)	**CANNOT ESTABLISH**	recorded read is attributed by declaration, not verified — neither establishes real access nor non-access; precautionary inclusion is the controller's decision, not a finding
BROAD-path-completeness — orders record O-0417 was not accessed through key k_7f3a during the window through ANY path (not only the path logger-orders observes)	**CANNOT ESTABLISH**	path completeness — exhaustiveness of paths available to the exposed credential not demonstrated (declared key scope only)

1 — What the submitted log records (record level)

Proposition	State	Basis
no read of orders record O-0417 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	ESTABLISHED — documentary. The submitted orders log contains no read of O-0417 attributed to the exposed key. A fact about the LOG'S CONTENTS, read as submitted; it does NOT, on its own, establish real-world non-access — that is R-A-world, where the declared anchor and the undemonstrated capture bite.
no read of invoices record I-0988 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	ESTABLISHED — documentary. The submitted invoices log contains no entry reading I-0988 attributed to the key. A fact about the log's contents; the coverage gap that defeats the world-level claim (R-B-world) does not change what the log, as submitted, contains.

no row-level read of customers record C-5521 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	ESTABLISHED — documentary. The submitted customers log records only a CALL-level query (/v1/customers/search {postcode_prefix:'29'} at 01:40, no row ids) attributed to the key — no row-level read of C-5521. A fact about the log's contents; whether C-5521 was returned by that query is the world-level question (R-C-world).
no read of addresses record A-7730 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**ESTABLISHED**	ESTABLISHED — documentary. The submitted addresses log contains no read of A-7730 attributed to the key. A fact about the log's contents; that this log is unchained and db-admin-writable is what defeats the world-level claim (R-D-world), not the record-level reading.
no read of invoices record I-3301 attributed to key k_7f3a is recorded in the submitted access log during the exposure window	**CONTRADICTED**	CONTRADICTED — bounded to the submitted record. The access log contains a read of I-3301 at 03:05 attributed to the key's prefix (a row-level entry on the hash-chained, anchored invoices log), so the claim that the log is silent on such a read is false. This is a claim about the RECORD, not the world. The attribution — that the read carries the api-key:k_7f3a prefix — is DECLARED, not independently verified, exactly as it is for R-A–R-D's ABSENCE of a read; the finding holds R-E's attribution to the same bar, not a lower one. What is contradicted is only that the log is silent; whether a real access occurred under that attribution is R-E-world.

Limitations

R-A-world — *orders record O-0417 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)* → **CANNOT ESTABLISH**

(DECLARED, not supported — structure present, but anchor only simulated and capture not demonstrated (verified anchor + attribution are necessary, not sufficient)). STRONGEST record, but DECLARED not supported. The logging STRUCTURE is present: logger-orders reported alive across the whole window, the orders log is hash-chained, contiguous and anchored with the config hash bound in, and there is no entry by the key (R-A-record). It stops short on two counts. (1) The anchor signatures and RFC-3161 tokens are simulated (a synthetic bundle), so the chain's trustworthiness currently rests on DECLARATION, not on an independently verifiable signature. (2) A continuous heartbeat shows the logger was UP, not that it CAPTURED every relevant access on this path (liveness is not capture). Independently verified anchor signatures, timestamp evidence and actor attribution would remove the identified integrity and attribution limitations; establishing the narrow non-access proposition would ADDITIONALLY require adequate evidence that all relevant accesses through the specified path were captured during the window — necessary conditions, not automatically sufficient. Reason class: declared (non-verifiable) integrity anchor + capture not demonstrated — distinct from R-D, where the integrity STRUCTURE is absent

R-B-world — *invoices record I-0988 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)* → **CANNOT ESTABLISH** (coverage gap — logger-invoices silent 02:09–02:55 (45 beats absent)). Fails on COVERAGE. logger-invoices has a gap — last beat 02:09:00, next 02:55:00 (2760s, 45 beats absent); the available heartbeat evidence does not establish continuous logger operation across that window. The submitted log records a read attributed to the key on the invoices table (I-3301 at 03:05, after the gap — tested as R-E), so the table was in reach. An access to I-0988 during the gap might not have been recorded: absence of a log entry over a window whose coverage is undemonstrated is not an observation of non-access. Reason class: coverage gap

R-C-world — *customers record C-5521 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)* → **CANNOT ESTABLISH** (object granularity — call-level logging over a query that matches the record). Fails on OBJECT granularity. customers is logged at CALL level (logcfg-v7), not row level. The key called /v1/customers/search with {postcode_prefix:'29'} at 01:40, recorded without row ids. C-5521's postcode is 29649 — inside that query. The log records that a matching query ran but not which rows it returned, so it cannot rule C-5521 in or out. Reason class: non-row-level logging over a matching call

R-D-world — *addresses record A-7730 was not accessed through key k_7f3a during the exposure window (real-world non-access, via the logged path, under the declared key scope)* → **CANNOT ESTABLISH** (integrity structure absent — log unchained, store writable by db-admin). Fails on INTEGRITY structure. The addresses table has log_chain 'none' and its log store sits on the data cluster, writable by db-admin. There is no entry by the key (R-D-record) — but a missing entry in a log that the privileged actor can edit carries no weight: the log cannot be relied on to have captured, or to still contain, the access it would need to show. Reason class: no tamper-evident log (structure absent) — distinct from R-A, where the structure is present but the anchor is only declared

R-E-world — *invoices record I-3301 was not accessed through key k_7f3a during the exposure window (real access, not merely the recorded entry)* → **CANNOT ESTABLISH** (recorded read is attributed by declaration, not verified — neither establishes real access nor non-access; precautionary inclusion is the controller's decision, not a finding). CANNOT ESTABLISH. The recorded read (R-E-record) is attributed by DECLARATION, not independently verified, so its presence does not establish that a real access by this key occurred — and, symmetrically, it does not establish non-access either. At the world level R-E is therefore as unestablished as R-A–R-D. A controller MAY, as a precautionary DECISION, treat R-E as potentially affected on the strength of the recorded read: the burden sits with the party asserting 'unaffected' (Art. 5(2) / 33(5)), so over-including a record is the safer error. That is a decision rule, NOT an evidentiary finding — the precaution does not convert a declared attribution into established fact. FactNotebook states the establishability; the inclusion decision stays with the controller.

BROAD-path-completeness — *orders record O-0417 was not accessed through key k_7f3a during the window through ANY path (not only the path logger-orders observes)* → **CANNOT ESTABLISH** (path completeness — exhaustiveness of paths available to the exposed credential not demonstrated (declared key scope only)). Fails on PATH COMPLETENESS. Even for the strongest record (R-A), 'not accessed' via one logger establishes only 'not accessed via THAT path'. The presence of other actors accessing these tables (svc-export reads orders & addresses, svc-report reads orders) shows that access is not limited to the exposed key; it does NOT, by itself,

establish additional technical paths, nor demonstrate that the exposed key could use them (those actors may traverse the same logged surface). What the submitted evidence does not establish is that ALL paths available to the exposed credential — including any delegation or session-derivation mechanism — were covered by the declared logging boundary. That completeness is declared in incident.json (key scope), not evidenced. Keep conditions 1-3 for the NARROW per-path proposition; path completeness is a fourth condition the BROAD proposition requires. Leaving it out of scope does not make the broad claim true — it silently narrows the proposition while the label ('not accessed') still reads broad

Separation of authority

Observation instruments determine what they observe and propose. Admission determines what may enter the reconstruction record. FactNotebook determines what that admitted record permits us to establish about the proposition. An instrument may search, select, transform or directly acquire observations according to its own declared procedure; those operations remain attributable properties of the instrument, not hidden properties of the reconstruction engine.

recorded behaviour → **observation** (attributed to its instrument) · declared risk / rule classifications → **the evaluator's / authority's policy** (never engine facts) · the consequential judgment → **the accountable authority**, outside this reconstruction.

Re-derivability

Two levels are preserved. **The finding** re-derives: re-running the reconstruction against the admitted observations reproduces the reconstruction identifier `FN-8c66ed94f110fa24` and the states in this document — with no re-run of the probabilistic extraction. **The admitted record's boundary** is separately inspectable via the candidate admission register (Annexes A/B) and the acquisition boundary (Annex C). *Version the instrument · trace the extraction · freeze the admitted record · replay the reconstruction.*

References

Each observation points to a citable source a third party can re-walk; the status records how it was verified.

Case-2 synthetic bundle (controller-supplied), integrity-verified — factnotebook_case2_v2: incident.json, repo_events.json, records_in_scope.csv, access_log.jsonl, log_anchors.json, logger_heartbeat.csv, logging_config.json (logcfg-v7), storage_attestation.json; + assess.py, build_evidence.py, README.md — all under the manifest · *PRIMARY — MANIFEST.sha256 recomputed here, 11/11 files match; facts re-derived independently of assess.py* · `synthetic pack supplied by Piotr Reder (case 2, v2); no real organisation, person or system`

Exposure window — start = commit_pushed 2026-09-03T00:00:00Z, end = key_revoked 2026-09-03T08:05:00Z · *PRIMARY — from repo_events.json, not assumed* · `evidence/repo_events.json; cross-checked against incident.json exposure_window_utc`

Coverage (R-B) — re-derived — logger-invoices: 441 alive beats in window, gap 02:09:00 -> 02:55:00 (45 beats absent); the other three loggers: full coverage, max gap 60s · *PRIMARY — recomputed from logger_heartbeat.csv* · `evidence/logger_heartbeat.csv`

Object granularity (R-C) — re-derived — customers logging_level 'call'; key call /v1/customers/search {postcode_prefix:'29'} at 01:40, row_ids null; C-5521 postcode 29649 starts with '29' · *PRIMARY — from access_log.jsonl + logging_config.json* · `evidence/access_log.jsonl; evidence/logging_config.json`

Integrity (R-D) / recorded access (R-E) — re-derived — addresses log_chain 'none', store writable by db-admin; invoices key read of I-3301 at 03:05 (row-level, on the chained+anchored log) · *PRIMARY — from logging_config.json + access_log.jsonl* · `evidence/logging_config.json; evidence/access_log.jsonl`

Path completeness (BROAD) — the declared-not-evidenced boundary — key_scope declared read-only with the api-key:k_7f3a prefix; access_log shows reads by other actors (svc-export -> orders & addresses, svc-report -> orders). The bundle shows access by other actors, but does not establish distinct technical paths or demonstrate that every path available to the exposed credential was observed · *OPEN — declared in incident.json, not evidenced by the bundle* · `evidence/incident.json (key_scope); evidence/access_log.jsonl (other actors)`

Observation register (obs -> provenance -> integrity -> probative role) — Observation count (6) is not proposition count (11). RECORD level (what the log contains, read as submitted): four observations each ESTABLISH a record-level silence (R-A/B/C/D-record), and the single recorded read of I-3301 CONTRADICTS R-E-record. WORLD level (real-world non-access): the five R-*-world propositions have NO establishing observation -> CANNOT ESTABLISH, each for its named reason (declared-anchor+capture / coverage / object-granularity / integrity-structure / declared-attribution); the recorded read of I-3301 leaves R-E-world unestablished too. BROAD (path completeness) also has no establishing observation (the sixth

observation documents the other-actor path facts without binding). The surface is SUBMITTED by the controller (synthetic): integrity is verified, independence and completeness are NOT — which is why no world-level non-access is established and the broad claim CANNOT ESTABLISH · *ADMISSION — six direct observations from one integrity-verified synthetic bundle drive eleven propositions (each record tested at two levels); no interpretive candidate, admitted without a separate interpretive admission step* · `per-proposition detail in the sections / limitations above`

Frozen reconstruction manifest (the inputs a replay fixes) — reconstruction_id FN-8c66ed94f110fa24 · observation_model FN-PIOTR-CASE2-001 hash sha256:1ac82995fa7417f2d0cc756ac85e890674fa55d7d92ae8a7b59655491b258899 · observation_set hash sha256:8d29be6e0f1fdffc356ee8ecc9955c8d665e50d4d8a7c9e30e108b86ecca23f7 (6 observations: o_1590b50510a54dd7, o_4bd93f802edbac6b, o_b779a74760231048, o_c491c45999abd2c7, o_c5dbded8e89015b9, o_c91a28a6f6e00843) · rule versions fn-reconstruction-engine-v1 / fn-rule-pattern-v1 / fn-composition-v1 / fn-temporal-v1 / fn-canon-v1. Internal determinism: re-running against this frozen bundle reproduces the identifier and the eleven proposition outcomes; the engine is not public, so independent verification is at the OBSERVATION level — the synthetic bundle can be re-hashed and its facts re-derived by anyone · *FROZEN — mirrors Annex D binding* · `Annex D (binding)`

Annex A — Admitted observation set

The admitted half of the candidate admission register — each an accountable ADMIT decision (determiner and criterion below). The full register (admitted and rejected) is in the machine-readable finding.

Candidate	Observation	Primary	SHA-256	Determined by	Criterion
obs-R-A	record:R-A · log_no_attr_read_RA	evidence/access_log.jsonl (+ logger_heartbeat.csv, log_anchors.json, logging_config.json)	5126b28f0605	FactNotebook	read from the integrity-verified bundle (MANIFEST.sha256 11/11); no orders read by the key -> R-A-record
obs-R-B	record:R-B · log_no_attr_read_RB	evidence/access_log.jsonl (+ logger_heartbeat.csv)	5126b28f0605	FactNotebook	no I-0988 read by the key recorded; heartbeat c5d9d779… shows the gap -> R-B-record
obs-R-C	record:R-C · log_no_attr_read_RC	evidence/access_log.jsonl (+ logging_config.json, records_in_scope.csv)	5126b28f0605	FactNotebook	only a call-level customers query by the key (no row ids); config d802bec9… level 'call' -> R-C-record
obs-R-D	record:R-D · log_no_attr_read_RD	evidence/access_log.jsonl (+ logging_config.json)	5126b28f0605	FactNotebook	no A-7730 read by the key; config d802bec9… addresses log_chain 'none' -> R-D-record
obs-R-E	record:R-E · attributed_read_recorded	evidence/access_log.jsonl	5126b28f0605	FactNotebook	a read of I-3301 at 03:05 attributed to the key is recorded -> contradicts R-E-record
obs-BROAD	record:BROAD · other_paths_read_table_completeness_declared	evidence/access_log.jsonl (+ incident.json)	5126b28f0605	FactNotebook	other actors read these tables; incident.json 3d6dc061… declares key scope -> BROAD path completeness

Annex B — Non-admitted candidate register

Every candidate was admitted: none rejected, none left unresolved.

Annex C — Acquisition / selection boundary

Records were constituted and supplied by the controller (synthetic) — Piotr Reder's case-2 pack, submitted to FactNotebook as one bundle of logs, heartbeats, anchors, config and incident metadata, and admitted without selective exclusion of candidate observations — all extracted candidates were admitted. This finding is bounded to that submitted surface; the absence of internal selection does not establish the submission's completeness relative to the represented system, nor whether an adverse class of observations was withheld — that completeness is not established.

Surface constituted / submitted by	the controller (synthetic) — Piotr Reder's case-2 pack, submitted to FactNotebook as one bundle of logs, heartbeats, anchors, config and incident metadata
Submission form	one integrity-sealed bundle (MANIFEST.sha256 over eight evidence files + assess.py + build_evidence.py + README.md = 11); 11/11 digests recomputed here and matched
Candidate extraction instrument	FactNotebook re-reads the bundle and re-derives every driving fact independently (heartbeat coverage per logger, the call-level query / postcode match, the chain structure, the recorded read). It does NOT re-run the controller's systems and does NOT re-use the controller's assess.py to reach its states
Admission authority / mechanism	FactNotebook (proposer). Integrity (SHA-256 over the whole bundle) is kept distinct from independence and from completeness: the bundle is the controller's own, so its completeness and independence are declared, not established — a submitted surface is never read as the world. Not an independent audit of the controller
Reconstruction engine	camsva 0.9.0
Surface type	submitted
Acquisition mode	submitted bundle — no candidate selection
Period	exposure window 2026-09-03T00:00:00Z .. 08:05:00Z (commit_pushed .. key_revoked)
Known limitations	Synthetic data, one scenario, five records, results per ROW (a data subject may appear in several tables; DS ids are carried but persons are not reconstructed here). Integrity is verified; independence and path completeness are NOT — path completeness is declared in incident.json; the bundle shows access by other actors, but does not establish distinct technical paths or demonstrate that every path available to the exposed credential was observed. Clock trust and attestor independence remain outside the independently verified scope. Logger capture completeness is explicitly considered (it is the second limit on R-A) but is not established by heartbeat evidence — liveness is not capture. No legal finding on notification is made.

Annex D — Policy & engine binding

Engine camsva 0.9.0 · Reconstruction ID `FN-8c66ed94f110fa24`

FactNotebook reconstructs what can be established from the surviving records — including what cannot be — and keeps the establishment separate from the judgment applied to it.